PIPELINE AUDIT AND HEALTH REPORT

Hospitality: pizza kitchen

Orders of pizza, pasta and salad are routed by type to three different stations. Pizzas share two ovens (shortest-queue), everything meets at the expo pass, and delivery drivers leave in pairs of orders or after 3 minutes. The ovens limit capacity at peak. Orders are lost when the order desk backlog is full.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 30 min) | seed 91 | 10 replications

HEALTH SCORE

87 /100

Grade B

Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 1 | Medium 1 | Low 1 | Info 2

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
27.5 units/h	215 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
6.9 units	900 s	oven_1

Executive summary

The line keeps up with demand (0.00% lost, sources never throttled): throughput equals the arrival rate, so nothing is binding right now. The busiest station is 'oven_1' at 91% busy; that is where capacity runs out first as demand grows.

What to fix first

- High (oven_1): Add parallel capacity, speed the station up, or reduce downtime.
- Medium (oven_2): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.
- Low (run length): Aim for at least 30x the cycle time so averages settle.

Best measured improvement

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Capacity	**oven_1**	The bottleneck is 95% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime.	-8

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Medium	Reliability	**oven_2**	Down 13% of the time (1 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Low	Model	**run length**	The measured window (7.5 h) is only 30x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**oven_1**	Highest utilisation in the line (95%); the line currently keeps up with demand. Action: This is where capacity runs out first as demand grows.	
Info	Reliability	**dependencies**	3 stations lie on every path (serial structure), but none has a failure model. Action: Fine for an ideal analysis; add failure data to find out how fragile this is.	

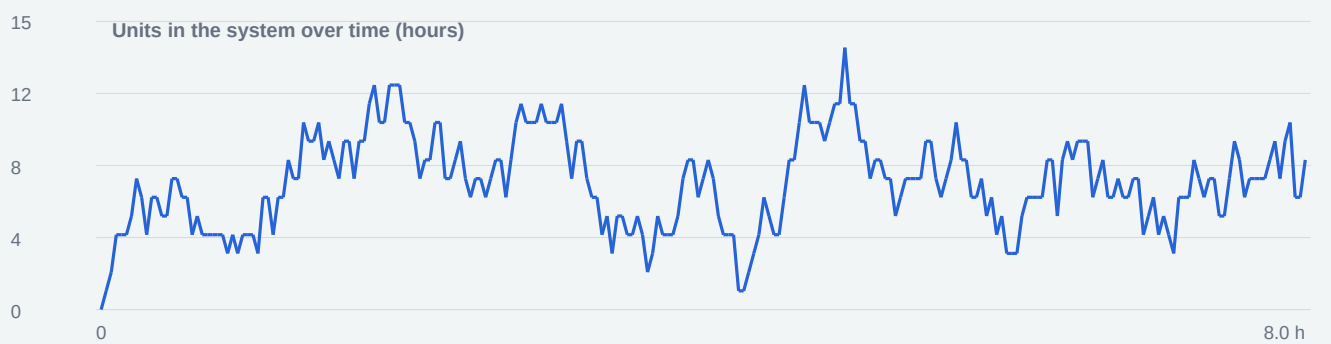
Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
orders	source	-	-	-	-	-
order_desk	machine		34.8	0.0	0.0	0.4 / 4
pizza_prep	machine		42.4	0.0	0.0	0.5 / 3
oven_1 bottleneck	machine		91.2	0.0	4.3	1.8 / 4
oven_2	machine		55.0	0.0	13.2	1.2 / 3
pasta_station	machine		85.2	0.0	0.0	1.9 / 4
salad_station	machine		11.9	0.0	0.0	0.1 / 2
pass	machine		23.9	0.0	0.0	0.3 / 2
dispatch	batcher		9.0	0.0	0.0	0.6 / 3
delivered	sink	-	-	-	-	-

Work in process over time



Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	27.11 units/h	+- 0.85	1.19	3.1%
Average WIP	6.54 units	+- 0.54	0.75	8.2%
Cycle time	845.96 s	+- 50.45	70.53	6.0%
Arrivals lost	0.56 %	+- 1.26	1.76	226.2%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
oven_1	360 min	15.0 min	96.0%	4.3%	2
oven_2	360 min	15.0 min	96.0%	13.2%	1

Single points of failure

All units must pass through: order_desk, pass, dispatch. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 220 = completed 215 + scrapped 0 + lost 0 + in system 5
Little's Law	PASS	L = 6.90, lambda x W = 6.87 (gap 0.5%)
Run length	WARN	window = 30 x cycle time (30x or more recommended)
Warm-up	PASS	first 30 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 26.9 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
oven_1	+2 buffer slots	26.9	**+0.0**	+0.0%
oven_1	10% faster processing	26.9	**+0.0**	+0.0%
oven_1	20% shorter repairs	26.9	**+0.0**	+0.0%
pasta_station	+2 buffer slots	26.9	**+0.0**	+0.0%
pasta_station	10% faster processing	26.9	**+0.0**	+0.0%
oven_2	+2 buffer slots	26.9	**+0.0**	+0.0%
oven_2	20% shorter repairs	26.9	**+0.0**	+0.0%
oven_2	10% faster processing	26.9	**-0.0**	-0.2%

No single local change moves throughput by more than 1%: the line is demand-limited or constrained elsewhere. Try raising the arrival rate, or look at the system-wide structure.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
orders	source	interarrival=exponential(130), mix={"pasta":0.2,"pizza":0.65,"salad":0.15}
order_desk	machine	capacity=8, on_full=drop, process=normal(45,10)
pizza_prep	machine	capacity=6, process=normal(90,20)
oven_1	machine	capacity=3, mtbf=21600, mttr=900, process=normal(300,40)
oven_2	machine	capacity=3, mtbf=21600, mttr=900, process=normal(300,40)
pasta_station	machine	capacity=4, process=normal(480,90)
salad_station	machine	capacity=4, process=normal(120,30)
pass	machine	capacity=6, process=normal(30,8)
dispatch	batcher	batch_size=2, capacity=6, process=20, timeout=180
delivered	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**oven_1**	The bottleneck is 95% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**oven_2**	Down 13% of the time (1 failures).	-3
Low	**run length**	The measured window (7.5 h) is only 30x the cycle time.	-2
	Total deductions		**-13**
	Health score		**87**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.